Finding and Fixing Related-Rates and Optimization Setup Errors
Answer Key

AP Calculus AB/BC

Quick Answers

1. 942.48
2. (b) corrected dV/dt = 24, —
3. (b) corrected x = 15, —
4. (b) corrected dy/dt = $-\frac{5}{6}$, —
5. (b) critical values = $\frac{10}{3}$, 10, —
6. (b) critical values = -3, 3, (c) minimum value = 6, —
7. (b) corrected ds/dt = $\frac{8}{3}$, —
8. (b) corrected S(r) = $2\pi r^2 + \frac{2000}{r}$, (c) minimizing radius = 5.42, —

What is verified

10 of 18 answers machine-checked against SymPy, across 8 problems.
— marks an answer only you can judge — problems 2, 3, 4, 5, 6, 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: AP Calculus AB/BC
CHA-3.D – *problems 1, 2, 4, 7*
FUN-4.B – *problems 3, 5, 8*
FUN-4.C – *problem 6*
Difficulty 1–5 of 5 across 18 tagged checks.

Common wrong answers

1. *If they got 314.16: differentiated the volume formula but never multiplied by dr/dt*
2. *If they got 48: dropped the ds/dt factor when differentiating V = s^3*
6. *If they got -6: evaluated f at the critical value outside the domain*
7. *If they got 1.6: used the walker's distance in place of the lamp-to-tip distance*
8. *If they got 4.3: minimized the halved surface area 2*pi*r^2 + 1000/r*

1. Spherical balloon, $\frac{dr}{dt} = 3$ cm/s. Find $\frac{dV}{dt}$ when $r = 5$ cm.

Solution. Differentiate $V = \frac{4}{3}\pi r^3$ with respect to *time*, so the chain rule contributes $\frac{dr}{dt}$:

$$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt} = 4\pi(5)^2(3) = 300\pi.$$

$$\boxed{\frac{dV}{dt} \approx 942.48}$$

Units: cubic centimetres per second.

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2. Cube, $\frac{ds}{dt} = 0.5$ cm/s, $s = 4$ cm. The student wrote $\frac{dV}{dt} = 3s^2 = 48$. (a) Name the error. (b) Corrected $\frac{dV}{dt}$.

Solution (a) — instructor-judged. The student differentiated $V = s^3$ with respect to s, not with respect to t. Differentiating with respect to time attaches the chain-rule factor $\frac{ds}{dt}$:

$$\frac{dV}{dt} = 3s^2 \frac{ds}{dt}.$$

A response earns full credit if it identifies the missing $\frac{ds}{dt}$; a response that only says "the arithmetic is wrong" does not.

Solution (b). $\frac{dV}{dt} = 3(4)^2(0.5) = 48(0.5)$.

$$\boxed{\frac{dV}{dt} = 24}$$

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3. Pen against a barn wall, 60 m of fence, $2x + y = 60$, $A = x(60 - 2x)$. (a) What went wrong in the last step? (b) Corrected x.

Solution (a) — instructor-judged. The student set the *area* equal to zero instead of its *derivative*. $A = 0$ locates the two ways to build a pen with no area at all ($x = 0$ and $x = 30$); the largest area sits where $A'(x) = 0$. Credit any wording that separates $A = 0$ from $A'(x) = 0$.

Solution (b). Expand and differentiate:

$$A(x) = 60x - 2x^2, \qquad A'(x) = 60 - 4x = 0.$$

$$\boxed{x = 15}$$

Then $y = 60 - 2(15) = 30$ and the maximum area is 450 square metres, larger than the 0 the student's answer produces — a useful sanity check.

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4. Ladder, $x^2 + y^2 = 169$, $\frac{dx}{dt} = 2$ ft/s, $x = 5$, $y = 12$. The student reported $\frac{dy}{dt} = \frac{5}{6}$. (a) Describe the error. (b) Corrected $\frac{dy}{dt}$.

Solution (a) — instructor-judged. Solving $2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$ gives $\frac{dy}{dt} = -\frac{x}{y} \cdot \frac{dx}{dt}$, and the student dropped the minus sign. The top of the ladder is sliding *down*, so $\frac{dy}{dt}$ must be negative. Credit either the algebraic or the physical argument, so long as the response states that the rate is negative.

Solution (b). $\frac{dy}{dt} = -\frac{5}{12}(2)$.

$$\boxed{\frac{dy}{dt} = -\frac{5}{6}}$$

That is about -0.83 feet per second.

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5. Open box from a 20 by 20 sheet, corner squares of side x. The student wrote $V = x(20-x)^2$. (a) Which factor is wrong? (b) Critical values. (c) Which one maximizes V?

Solution (a) — instructor-judged. Each 20 cm edge loses a square at *both* ends, so the base measures $(20-2x)$ by $(20-2x)$:

$$V(x) = x(20-2x)^2.$$

Credit any wording that explains why the edge loses $2x$ rather than x.

Solution (b). Differentiating gives $V'(x) = (20-2x)^2 + x \cdot 2(20-2x)(-2) = (20-2x)(20-6x)$. Setting each factor to zero:

$$\boxed{x = \frac{10}{3} \text{ or } x = 10}$$

Solution (c) — instructor-judged. At $x = 10$ the base has collapsed to zero width, so $V = 0$ — the smallest volume possible, not the largest. At $x = \frac{10}{3}$ the volume is positive, so that is the maximizer. A first-derivative sign chart, a second-derivative test, or a direct evaluation at both values all earn full credit, provided $x = 10$ is rejected for a stated reason.

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6. Minimize $f(x) = x + \frac{9}{x}$ on $x > 0$. The student reported the minimum at $x = -3$. (a) Name the error. (b) Both solutions of $f'(x) = 0$. (c) The minimum value.

Solution (a) — instructor-judged. Two things went unchecked: the stated domain is $x > 0$, which rules $x = -3$ out entirely, and no test was applied to decide which critical value is a minimum. Since $f''(x) = \frac{18}{x^3} > 0$ for $x > 0$, the curve is concave up there and the critical value inside the domain is a minimum. Credit the domain argument, a second-derivative test, or a sign chart.

Solution (b). $f'(x) = 1 - \frac{9}{x^2} = 0$ gives $x^2 = 9$.

$$\boxed{x = -3 \text{ or } x = 3}$$

Solution (c). Only $x = 3$ lies in the domain, and $f(3) = 3 + \frac{9}{3}$.

$$\boxed{f(3) = 6}$$

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7. Lamp 15 ft tall, walker 6 ft tall moving at 4 ft/s; x is the walker's distance from the pole, s the shadow length. The student wrote $\frac{6}{15} = \frac{s}{x}$ and got 1.6 ft/s. (a) Describe the error. (b) Corrected $\frac{ds}{dt}$.

Solution (a) — instructor-judged. The two similar triangles share the shadow *tip*, not the walker's position: the large triangle has base $s + x$ (pole to tip) and height 15; the small one has base s and height 6. The correct proportion is therefore

$$\frac{6}{15} = \frac{s}{s+x},$$

not $\frac{6}{15} = \frac{s}{x}$. Credit any wording that puts $s + x$ in the denominator.

Solution (b). Cross-multiplying, $6(s+x) = 15s$, so $6x = 9s$ and $s = \frac{2}{3}x$. Differentiating, $\frac{ds}{dt} = \frac{2}{3} \cdot 4$.

$$\boxed{\frac{ds}{dt} = \frac{8}{3}}$$

About 2.67 feet per second — faster than the student's figure, as it must be, since the student's proportion understates the shadow.

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8. Closed can of volume 1000 cm^3. The student used $S = 2\pi r^2 + \pi r h$ and got $r \approx 4.30$. (a) Identify the error. (b) Corrected $S(r)$. (c) Minimizing radius.

Solution (a) — instructor-judged. A closed cylinder has two circular ends *and* a lateral surface that unrolls into a rectangle of width $2\pi r$, so the side contributes $2\pi r h$, not $\pi r h$. The student's objective is missing a factor of 2 on the lateral term. Credit any wording that names that missing factor.

Solution (b). With $h = \frac{1000}{\pi r^2}$,

$$S(r) = 2\pi r^2 + 2\pi r \cdot \frac{1000}{\pi r^2}.$$

$$\boxed{S(r) = 2\pi r^2 + \frac{2000}{r}}$$

Solution (c). $S'(r) = 4\pi r - \frac{2000}{r^2} = 0$ gives $4\pi r^3 = 2000$, so $r^3 = \frac{500}{\pi}$.

$$\boxed{r \approx 5.42}$$

The student's halved objective produces 4.30 instead, which is exactly the misconception this problem targets.